

EXPERIMENT-14

14. | Write a code using RAW sockets to implement packet sniffing.

packet_sniffer.py

```
import socket

HOST = socket.gethostbyname(socket.gethostname())

sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)

sniffer.bind((HOST, 0)) sniffer.setsockopt(socket.IPPROTO_IP,
socket.IP_HDRINCL, 1)

print("Packet Sniffer Started")

print("Listening on:", HOST) print("Press
Ctrl+C to stop")

try:

    while True:

        data, address = sniffer.recvfrom(65535)

print("Source IP:", address[0])    print("Packet
Size:", len(data), "bytes")    print("-" * 40)

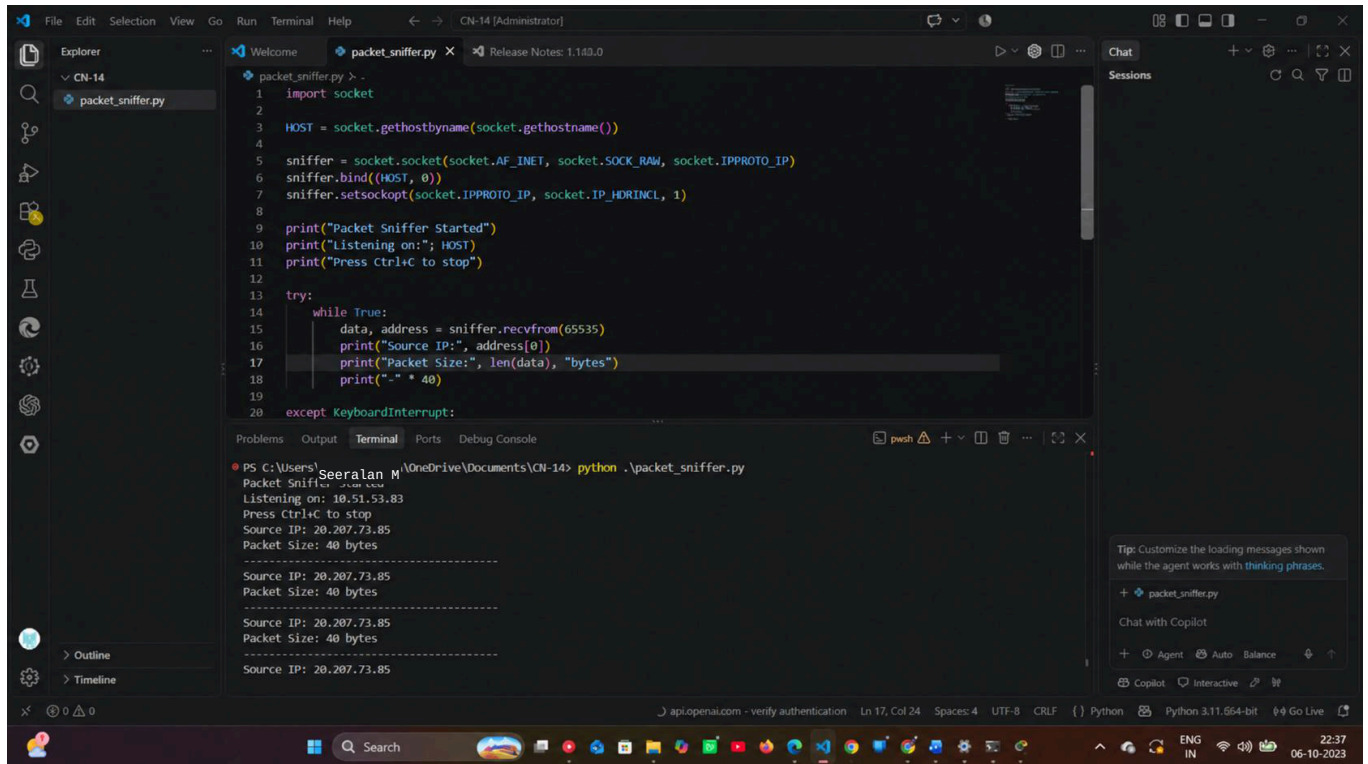
except KeyboardInterrupt:

    print("\nPacket Sniffer Stopped")

finally:

    sniffer.close()
```

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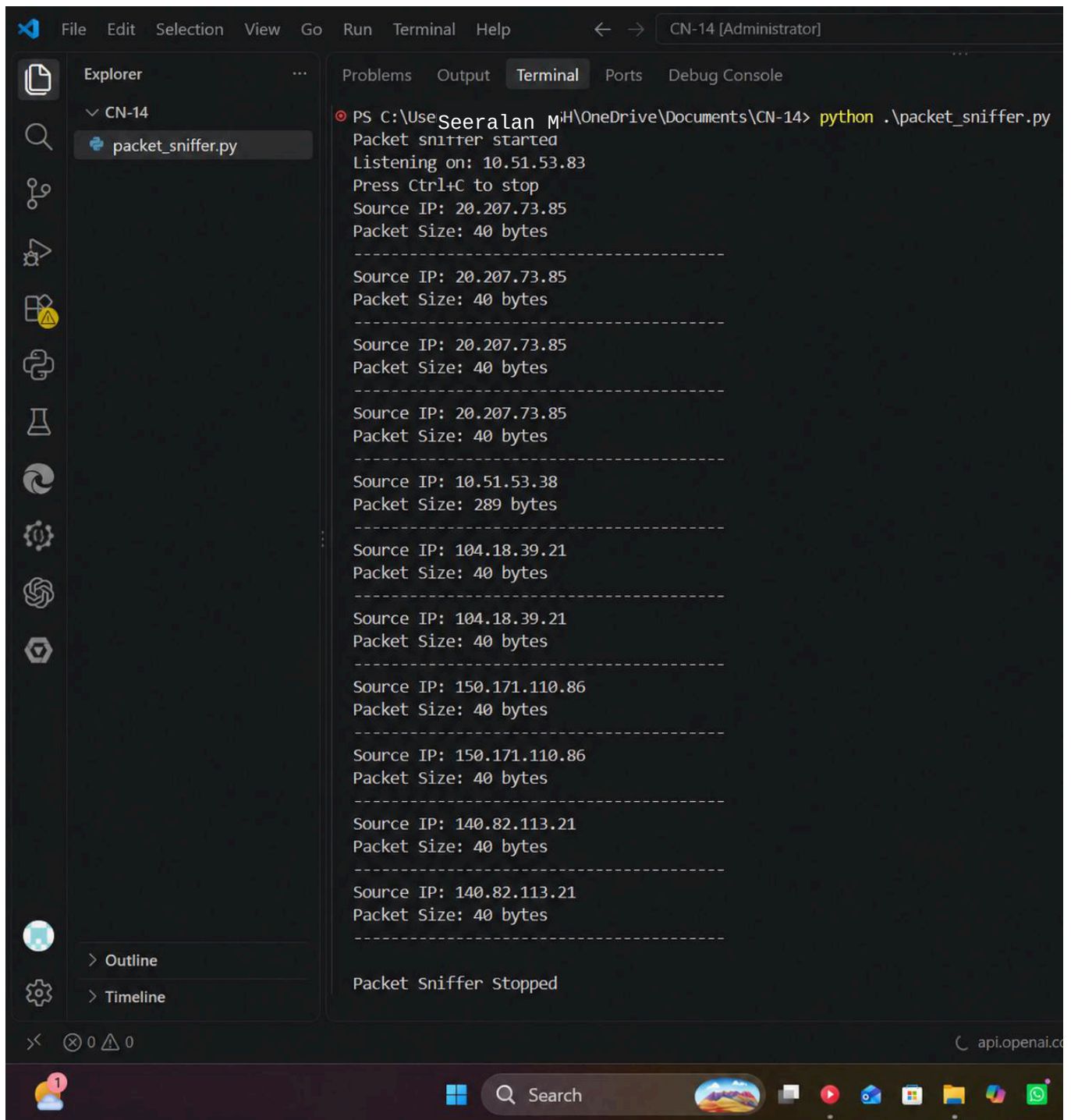


The screenshot shows a Windows IDE with a dark theme. The Explorer pane on the left shows a project named 'CN-14' with a file 'packet_sniffer.py'. The main editor displays the code for 'packet_sniffer.py', which is a Python script using the 'socket' module to create a raw socket and listen for incoming packets. The script prints the source IP and packet size for each received packet. The Terminal pane at the bottom shows the command 'python .\packet_sniffer.py' being executed, and the output shows the script listening on 10.51.53.83 and receiving three packets from 20.207.73.85, each 40 bytes in size. The status bar at the bottom indicates the file is encoded in UTF-8, uses CRLF line endings, and is a Python 3.11.564-bit file.

```
1 import socket
2
3 HOST = socket.gethostbyname(socket.gethostname())
4
5 sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
6 sniffer.bind((HOST, 0))
7 sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
8
9 print("Packet Sniffer Started")
10 print("Listening on: ", HOST)
11 print("Press Ctrl+C to stop")
12
13 try:
14     while True:
15         data, address = sniffer.recvfrom(65535)
16         print("Source IP:", address[0])
17         print("Packet Size:", len(data), "bytes")
18         print("-" * 40)
19
20 except KeyboardInterrupt:
```

```
PS C:\Users\Seeralan M\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet Sniffer Started
Listening on: 10.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
```

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```
File Edit Selection View Go Run Terminal Help
CN-14 [Administrator]

Explorer
CN-14
packet_sniffer.py

Problems Output Terminal Ports Debug Console

PS C:\Users\Seeralan M\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet sniffer started
Listening on: 10.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 10.51.53.38
Packet Size: 289 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Packet Sniffer Stopped
```